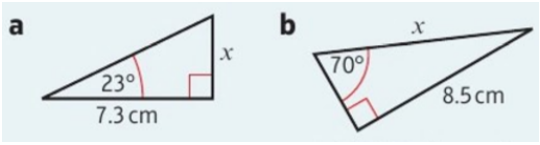
Trigonometric Ratios

Starters

Solution 2

1. Find the marked lengths.

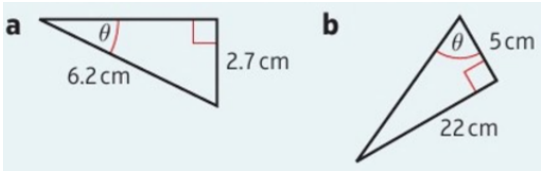


1(a,b)

$x = 7.3 \tan 23° \approx 3.10$ cm

$x = 8.5 / \sin 70° \approx 9.05$ cm

2. Find the marked angles.



2(a,b)

$\theta = \sin^{-1}(2.7/6.2) \approx 25.8°$

$\theta = \tan^{-1}(22/5) \approx 77.2°$

3. Given $f(x) = x^2 + 3x$, sketch each graph. All four are upward-opening parabolas.

a.

$y = f(x)$

a

$y = x(x+3)$

Roots -3, 0; vertex $(-1.5, -2.25)$.

b.

$y = f(x+2)$

b

$y = (x+2)(x+5)$

Roots -5, -2; y-intercept 10; vertex $(-3.5, -2.25)$.

c.

$y = f(x) - 3$

c

$y = x^2 + 3x - 3$

$x = \dfrac{-3 \pm \sqrt{21}}{2}$

y-intercept -3; vertex $(-1.5, -5.25)$.

d.

$y = f(\frac{1}{2}x)$

d

$y = \dfrac{1}{4}x(x+6)$

Roots -6, 0; vertex $(-3, -2.25)$.

Chapter Overview

Solution 3

1. Sine and cosine rules

Find missing sides and angles in non-right-angled triangles.

2. Areas of triangles

Use two sides and the included angle.

3. Trigonometric graphs

Sketch sine, cosine and tangent, then apply transformations.

RECAP: Right-Angled Trigonometry

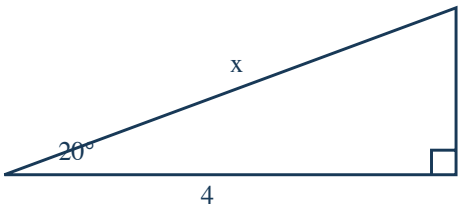
Solution 4

Meaning. What does $\sin\theta =$ opposite/hypotenuse mean?

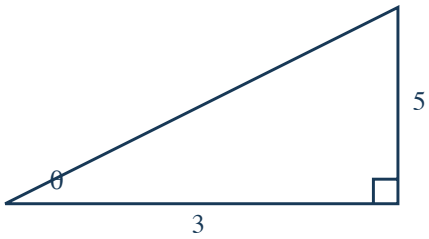
Meaning

For a given angle, sine gives the ratio of the opposite side to the hypotenuse. Similar right-angled triangles have the same ratio.

Find x.



Find θ.



x

$$\cos 20^\circ = \frac{4}{x}$$

$$x = \frac{4}{\cos 20^\circ} = 4.26 \text{ (3 s.f.)}$$

θ

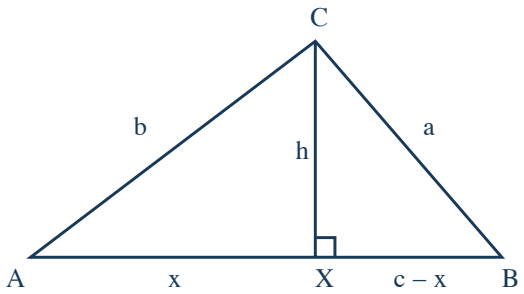
$$\tan\theta = \frac{5}{3}$$

$$\theta = \tan^{-1}(5/3) = 59.0^\circ$$

Write the ratio equation first. Then rearrange for a length, or apply the inverse trigonometric function for an angle. Use degree mode.

Proof of Cosine Rule

Solution 5



Use Pythagoras in the two right-angled triangles.

$$h^2 + x^2 = b^2$$
$$h^2 + (c - x)^2 = a^2$$

Drop a perpendicular CX to AB. Let AX=x and AB=c.

2. Eliminate h^2 and expand.

2

$$a^2 - b^2 = (c - x)^2 - x^2$$
$$a^2 = b^2 + c^2 - 2cx$$

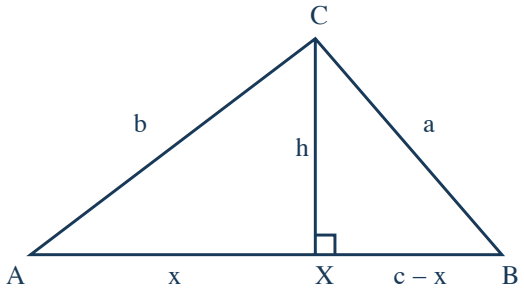
3. Use the angle at A.

3

$$x = b\cos A$$
$$a^2 = b^2 + c^2 - 2bc\cos A$$

Proof of Sine Rule

Solution 6



1. Express h in two ways.

1

$$h = b\sin A = a\sin B$$

The same altitude h belongs to both right-angled triangles.

2. Rearrange.

2

$$a\sin B = b\sin A$$
$$\frac{a}{\sin A} = \frac{b}{\sin B}$$

3. Repeat with another altitude.

3

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Just for your interest... Solution 7

Why does “cosine” contain the word “sine”?

Complementary angles add to 90°.

Supplementary angles add to 180°.

1. The acute angles of a right-angled triangle are 50° and 40°. Let x be the side opposite 40° and z the hypotenuse. Compare cosine of 50° with sine of 40°.

1

$$\cos 50^\circ = \frac{x}{z}$$

$$\sin 40^\circ = \frac{x}{z}$$

2. Generalise this relationship.

2

$$\cos\theta = \sin(90^\circ - \theta)$$

Cosine means the sine of the complementary angle.

OVERVIEW: Finding missing sides and angles Solution 8

Choose the rule from the information available. Lower-case side labels are opposite the matching capital-letter angles.

1. A known opposite side–angle pair and another side or angle.

1

Sine rule. When finding an angle, check for a possible second solution.

2. Two sides and the included angle; find the third side.

2

Cosine rule.

3. All three sides; find an angle.

3

Cosine rule.

4. Two sides and a non-included angle; find another side.

4

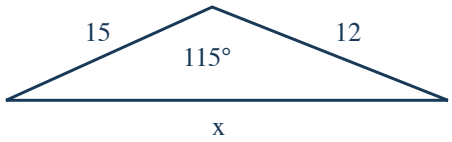
Sine rule to find an angle, angle sum to find the remaining angle, then sine rule again. Check the ambiguous case.

Cosine Rule

Solution 9

$a^2 = b^2 + c^2 - 2bc\cos A$

A is the angle between b and c, opposite a.



Find x.

x

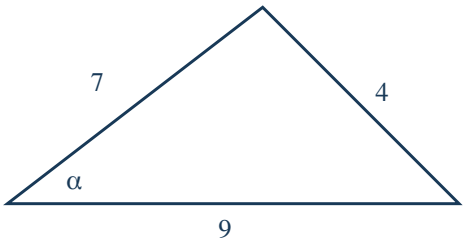
$x^2 = 15^2 + 12^2 - 2(15)(12)\cos 115°$

$x^2 = 521.14257\ldots$

$x = 22.83$ (4 s.f.)

Dealing with Missing Angles

Solution 10



Find α, opposite the side of length 4.

$4^2 = 7^2 + 9^2 - 2(7)(9)\cos\alpha$

$16 = 130 - 126\cos\alpha$

$\cos\alpha = \dfrac{114}{126}$

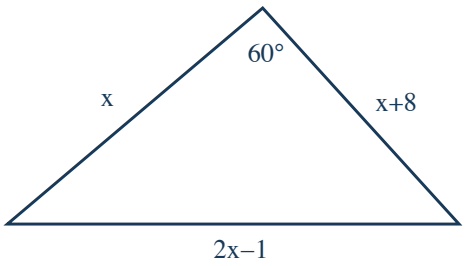
$\alpha = \cos^{-1}(114/126) = 25.2°$

Alternatively use $\cos A = (b^2 + c^2 - a^2)/(2bc)$. Only the product $2bc$ is multiplied by $\cos A$, not the entire right-hand side.

Harder Ones

Solution 11

1. Determine x.



1

$(2x-1)^2 = x^2 + (x+8)^2 - 2x(x+8)\cos 60°$

$3x^2 - 12x - 63 = 0$

$(x+3)(x-7) = 0$

Lengths must be positive, so x=7.

2. Coastguard station B is 8 km on a bearing of 060° from A. Ship C is 4.8 km on a bearing of 018° from A. Find BC.

2

$\angle BAC = 60° - 18° = 42°$

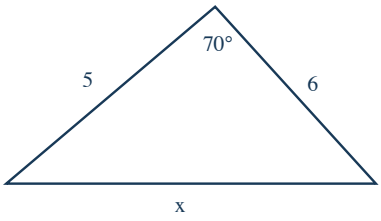
$BC^2 = 8^2 + 4.8^2 - 2(8)(4.8)\cos 42°$

$BC = 5.47 \text{ km (3 s.f.)}$

Test Your Understanding

Solution 12

1. Find x.

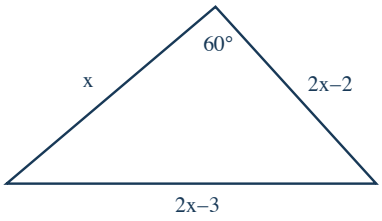


1

$x^2 = 5^2 + 6^2 - 60\cos 70°$

$x = 6.36 \text{ (3 s.f.)}$

2. Find x.



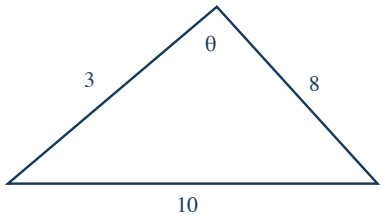
2

$(2x-3)^2 = x^2 + (2x-2)^2$
$\quad -2x(2x-2)\cos 60°$

$x^2 - 6x + 5 = 0$

x=1 or 5. Positive side lengths require x=5.

3. Find θ.



3

$\cos\theta = \dfrac{3^2 + 8^2 - 10^2}{2(3)(8)}$

$\theta = 124.2°$

An obtuse angle has a negative cosine.

Exercise 9A – Y1 pages 177-179 Solution 13

8. Triangle PQR has PQ=2 cm, PR=3 cm and QR=4 cm. Show that $\cos P = -\frac{1}{4}$.

8

$$\cos P = \frac{2^2 + 3^2 - 4^2}{2(2)(3)} = -\frac{1}{4}$$

11. The side lengths of a triangle are in the ratio 2:3:4. Calculate the largest angle.

11

The largest angle B is opposite the longest side, 4k.

$$\cos B = \frac{(2k)^2 + (3k)^2 - (4k)^2}{2(2k)(3k)}$$

$$= -\frac{1}{4}$$

$$B = 104.477\ldots^\circ \approx 104^\circ \text{ (3 s.f.)}$$

Exercise 9A – Y1 pages 177-179 Solution 14

17. Ships A, B and C lie in the same horizontal plane. B is 50 km due north of A. C is 70 km from A on a bearing of 020°.

a. Find BC to 3 significant figures.

a

$$BC^2 = 50^2 + 70^2 - 2(50)(70)\cos 20^\circ$$

$$BC = 28.673\ldots \approx 28.7 \text{ km}$$

b. Find the bearing of C from B.

b

$$\cos B = \frac{50^2 + BC^2 - 70^2}{2(50)(BC)}$$

$$B = 123.3867\ldots^\circ$$

$$\text{Bearing} = 180^\circ - B = 056.6^\circ$$

Extension

Solution 15

[STEP I 2009 Q4(i)] A triangle has sides p−q, p and p+q, where p>q>0. Its largest and smallest angles are α and β. Show that

$4(1-\cos\alpha)(1-\cos\beta)=\cos\alpha+\cos\beta$

1. Apply the cosine rule opposite the longest and shortest sides.

1

$$\cos\alpha=\frac{p-4q}{2(p-q)}$$

$$\cos\beta=\frac{p+4q}{2(p+q)}$$

2. Compare both sides of the identity.

2

$$1-\cos\alpha=\frac{p+2q}{2(p-q)}$$

$$1-\cos\beta=\frac{p-2q}{2(p+q)}$$

$$4(1-\cos\alpha)(1-\cos\beta)=\frac{p^2-4q^2}{p^2-q^2}$$

$$\cos\alpha+\cos\beta=\frac{p^2-4q^2}{p^2-q^2}$$

The Sine Rule

Solution 16

Compare side length divided by the sine of its opposite angle.

1. Opposite 85°: side 10; opposite 30°: side 5.02; opposite 65°: side 9.10.

1

$$\frac{10}{\sin 85^\circ}\approx 10.04$$

$$\frac{5.02}{\sin 30^\circ}=10.04$$

$$\frac{9.10}{\sin 65^\circ}\approx 10.04$$

$$\frac{a}{\sin A}=\frac{b}{\sin B}=\frac{c}{\sin C}$$

Use the sine rule when you know a complete opposite side–angle pair and one part of another pair. Small discrepancies above come from rounded side lengths.

Examples

Solution 17

1. In a triangle, side 8 is opposite 45° and side x is opposite 85°. Find x.

1

$$\frac{x}{\sin 85^\circ} = \frac{8}{\sin 45^\circ}$$

$$x = \frac{8 \sin 85^\circ}{\sin 45^\circ} = 11.27 \text{ (4 s.f.)}$$

2. In a triangle, side 8 is opposite 30° and side x is opposite 100°. Find x.

2

$$\frac{x}{\sin 100^\circ} = \frac{8}{\sin 30^\circ}$$

$$x = \frac{8 \sin 100^\circ}{\sin 30^\circ} = 15.76 \text{ (4 s.f.)}$$

Match each side to its opposite angle. The remaining angles are 50° in both examples.

Examples

Solution 18

For a missing angle it is convenient to use $\sin A/a = \sin B/b$.

3. Side 6 is opposite 85°, and side 5 is opposite θ. Find θ.

3

$$\sin\theta = \frac{5 \sin 85^\circ}{6}$$

$$\theta = 56.11^\circ$$

The supplementary angle is impossible because it and 85° would exceed 180°.

4. Side 10 is opposite 126°, and side 8 is opposite θ. Find θ.

4

$$\sin\theta = \frac{8 \sin 126^\circ}{10}$$

$$\theta = 40.33^\circ$$

The other angle must be acute since 126° is already obtuse.

Exercise 9B – Y1 pages 181-183 Solution 19

8. B is 6 km on a bearing of 020° from A. C is on a bearing of 055° from A and 120° from B. Find AC and BC.

Angles. Find the interior angles.

Angles

$$A = 55^\circ - 20^\circ = 35^\circ$$

$$B = 200^\circ - 120^\circ = 80^\circ$$

$$C = 65^\circ$$

a. Find AC.

a

$$AC = \frac{6 \sin 80^\circ}{\sin 65^\circ} = 6.52 \text{ km}$$

b. Find BC.

b

$$BC = \frac{6 \sin 35^\circ}{\sin 65^\circ} = 3.80 \text{ km}$$

Exercise 9B – Y1 pages 181-183 Solution 20

11. In triangle ABC, AB=x cm, BC=(4−x) cm, angle BAC=y and angle BCA=30°. Given $\sin y = 1/\sqrt{2}$, show that $x = 4(\sqrt{2} - 1)$.

1. Use the sine rule.

1

$$\frac{x}{\sin 30^\circ} = \frac{4 - x}{\sin y}$$

$$2x = \sqrt{2}(4 - x)$$

$$x = \frac{4}{\sqrt{2} + 1}$$

2. Rationalise the denominator.

2

$$x = \frac{4(\sqrt{2} - 1)}{(\sqrt{2} + 1)(\sqrt{2} - 1)}$$

$$x = 4(\sqrt{2} - 1)$$

Exercise 9B – Y1 pages 181-183 Solution 21

12. A surveyor measures the angles of elevation of the top of a building from two points 15 m apart, on level ground in a straight line towards the building. The angles are 40° at the farther point and 52° at the nearer point.

a. Find the height of the building.

b. State an assumption.

a

The triangle joining the two observation points and the top has angles 40°, 128° and 12°.

$$\text{Farther sloping distance} = \frac{15 \sin 128^\circ}{\sin 12^\circ}$$

$$h = \frac{15 \sin 128^\circ \sin 40^\circ}{\sin 12^\circ}$$

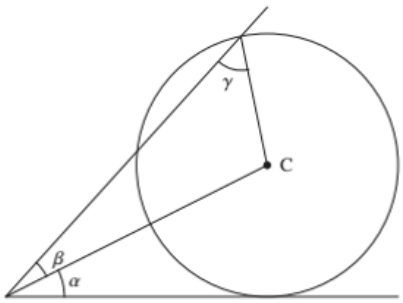
$$h = 36.5 \text{ m (3 s.f.)}$$

b

The angles of elevation are measured from ground level; the ground is horizontal and the building vertical.

Extension Solution 22

[MAT 2011 1E] The circle has centre C. Which relationship is correct?



A: $\cos\alpha = \sin(\beta + \gamma)$

B: $\sin\beta = \sin\alpha \sin\gamma$

C: $\sin\beta(1 - \cos\alpha) = \sin\gamma$

D: $\sin(\alpha + \beta) = \cos\gamma \sin\alpha$

Answer

Let the radius be 1 and the distance from the external point to C be x. Drop a perpendicular from C to the horizontal tangent.

$$1 = x \sin\alpha \quad \Rightarrow \quad x = 1/\sin\alpha$$

$$\frac{x}{\sin\gamma} = \frac{1}{\sin\beta}$$

$$\sin\beta = \sin\alpha \sin\gamma$$

Answer B.

The ‘Ambiguous Case’ — Solution 23

AB=4, AC=3 and angle ABC=44˚. Find the possible values of angle ACB.

1. Apply the sine rule.

2. Find both possible angles.

1

$$\frac{\sin C}{4} = \frac{\sin 44^\circ}{3}$$

$$\sin C = 0.92621\ldots$$

2

$$C = 67.9^\circ \quad \text{or} \quad 112.1^\circ$$

Both give a positive remaining angle, so both triangles are possible.

The ambiguous case: $\sin\theta = \sin(180^\circ - \theta)$. The calculator returns the principal inverse-sine value; the sine equation may also have a supplementary solution. Check the triangle and any stated conditions.

Test Your Understanding — Solution 24

A triangle has side 10 opposite an obtuse angle θ, and side 5 opposite 20˚. The remaining side is x.

1. Find θ.

2. Find x.

1

$$\sin\theta = \frac{10\sin 20^\circ}{5}$$

$$\theta = 180^\circ - 43.1602\ldots^\circ$$

$$= 136.8398\ldots^\circ$$

2

$$\text{Remaining angle} = 23.1602\ldots^\circ$$

$$x = \frac{5\sin 23.1602\ldots^\circ}{\sin 20^\circ}$$

$$x = 5.75 \text{ (3 s.f.)}$$

Exercise 9C – Y1 pages 184-185 Solution 25

4. Triangle ABC has AB=4 cm, BC=6 cm and angle ACB=36˚. Show that one possible value of angle ABC is 25.8˚ (3 s.f.), and use this case to find AC.

a. Find the relevant angle.

a

$$\sin A = \frac{6 \sin 36^\circ}{4}$$

$$A = 61.845\ldots^\circ \text{ or } 118.154\ldots^\circ$$

$$B = 180^\circ - 36^\circ - 118.154\ldots^\circ$$

$$B = 25.8^\circ \text{ (3 s.f.)}$$

b. Find AC using unrounded angles.

b

$$AC = \frac{4 \sin 25.845\ldots^\circ}{\sin 36^\circ}$$

$$AC = 2.97 \text{ cm (3 s.f.)}$$

Exercise 9C – Y1 pages 184-185 Solution 26

5. Two triangles ABC have AB=4.5 cm, BC=6.8 cm and angle ACB=30˚. Find the largest angle in each triangle.

1. Find both possible values of A.

1

$$\sin A = \frac{6.8 \sin 30^\circ}{4.5}$$

$$A = 49.073\ldots^\circ \text{ or } 130.926\ldots^\circ$$

2. Identify the largest angles.

2

$$B = 180^\circ - 30^\circ - 49.073\ldots^\circ$$

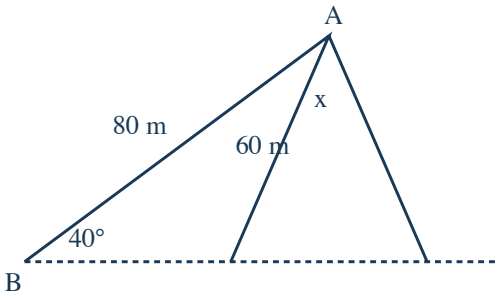
$$B = 101^\circ \text{ (3 s.f.)}$$

In the other triangle A is the largest angle:

$$A = 131^\circ \text{ (3 s.f.)}$$

Exercise 9C – Y1 pages 184-185 Solution 27

6. Crane arm AB is 80 m long and makes an angle of 40° with the horizontal at B. A wrecking ball hangs on a 60 m cable from A.



Find the angle x through which the cable rotates between the two positions where the ball is level with B.

a

$$h = 80\sin 40^\circ$$

$$\cos(x/2) = \frac{80\sin 40^\circ}{60}$$

$$x = 2\cos^{-1}\left(\frac{80\sin 40^\circ}{60}\right)$$

$$x = 62.0^\circ \text{ (3 s.f.)}$$

State a modelling assumption.

b

The cable is taut and has constant length; the ball swings symmetrically in a vertical plane.

Area of Non Right-Angled Triangles Solution 28

$$\text{Area} = \frac{1}{2}ab\sin C$$

C is the included angle between sides a and b.

Example. Find the area of a triangle with sides 3 cm and 7 cm and included angle 59°.

Example

$$\text{Area} = \frac{1}{2}(3)(7)\sin 59^\circ$$

$$= 9.00 \text{ cm}^2 \text{ (3 s.f.)}$$

Test Your Understanding

Solution 29

1. A triangle has sides x and x+3, included angle 30°, and area 10. Find x.

1

$$\frac{1}{2}x(x+3)\sin 30^\circ = 10$$

$$x^2 + 3x - 40 = 0$$

$$(x+8)(x-5) = 0$$

Positive lengths require x=5.

2. A triangle has sides 5 and 6, included obtuse angle θ, and area 10. Find θ.

2

$$15\sin\theta = 10$$

$$\theta = 180^\circ - \sin^{-1}(2/3)$$

$$\theta = 138.2^\circ$$

Exercise 9D – Y1 pages 186-187

Solution 30

4. Triangle ABC has BC=(x+2) cm, AC=x cm, angle BCA=150° and area 5 cm². Find x to 3 significant figures.

4

$$\frac{1}{2}x(x+2)\sin 150^\circ = 5$$

$$x^2 + 2x - 20 = 0$$

$$x = -1 \pm \sqrt{21}$$

Taking the positive root:

$$x = 3.58 \text{ (3 s.f.)}$$

Exercise 9D – Y1 pages 186-187 Solution 31

5. Triangle PQR has PQ=(x+2) cm, PR=(5−x) cm and angle QPR=30°. Its area is A cm².

a. Show that $A = \frac{1}{4}(10 + 3x - x^2)$.

b. Find the maximum area and the corresponding x.

a

$$A = \frac{1}{2}(x+2)(5-x)\sin 30^\circ$$

$$A = \frac{1}{4}(10 + 3x - x^2)$$

b

$$A = \frac{49}{16} - \frac{1}{4}\left(x - \frac{3}{2}\right)^2$$

The squared term is non-negative.

$$A_{\max} = \frac{49}{16}\ \text{cm}^2 \quad \text{at } x = \frac{3}{2}$$

Exercise 9D – Y1 pages 186-187 Solution 32

6. Triangle ABC has AB=x cm, AC=(5+x) cm, angle BAC=150° and area 3¾ cm².

a. Show that $x^2 + 5x - 15 = 0$.

b. Calculate x to 3 significant figures.

a

$$\frac{1}{2}x(5+x)\sin 150^\circ = \frac{15}{4}$$

$$x^2 + 5x - 15 = 0$$

b

$$x = \frac{-5 \pm \sqrt{85}}{2}$$

Reject the negative root.

$$x = 2.11 \text{ (3 s.f.)}$$

Sin or cosine rule?

Solution 33

Choose sine or cosine rule, and explain why. Check that the measurements can form a triangle.

1. Two sides are 3 and 5; their included angle is 100°. Find the opposite side.

1

Cosine rule: two sides and the included angle are known.

$a^2 = 3^2 + 5^2 - 30\cos 100^\circ$

2. Two angles are 40° and 45°, and one side is 3. Find another side.

2

Sine rule: first find the third angle, 95°, if needed, then match opposite pairs.

3. The three proposed side lengths are 2, 3 and 5. Find an angle.

3

These do not form a non-degenerate triangle: $2+3=5$. The cosine rule would give 180° opposite the longest side.

4. Two sides are 3 and 5, with 40° opposite side 3. Find the remaining side.

4

Use sine rule twice, checking whether an angle is possible.

$\sin A = \dfrac{5\sin 40^\circ}{3} > 1$

These measurements cannot form a triangle.

Using sine rule twice

Solution 34

Two sides of a triangle are 4 and 3. The angle opposite side 3 is 32°. Let the remaining side be x.

Cosine route. What equation would the cosine rule give?

Cosine route

$3^2 = x^2 + 4^2 - 8x\cos 32^\circ$

$x^2 - 8x\cos 32^\circ + 7 = 0$

This quadratic has two positive roots. The sine rule offers another route.

For side–side–angle information, there may be two triangles. A diagram alone should not be used to discard the other solution unless an angle condition is stated.

Using sine rule twice

Solution 35

Continue: side 3 is opposite 32°, side 4 is opposite A, and the remaining side is x.

1. Use the sine rule for A.

2. Find the remaining angle B.

1

$$\sin A = \frac{4 \sin 32^\circ}{3}$$

$$A = 44.9556\ldots^\circ \text{ or } 135.0444\ldots^\circ$$

2

$$B = 180^\circ - 32^\circ - A$$

$$B = 103.0444\ldots^\circ \text{ or } 12.9556\ldots^\circ$$

3. Use the sine rule again.

3

$$x = \frac{3 \sin B}{\sin 32^\circ}$$

$$x = 5.52 \text{ or } 1.27 \text{ (3 s.f.)}$$

If A is specified to be acute, choose x=5.52.

Test Your Understanding

Solution 36

1. A triangle has side 9 opposite 61° and another side 10. Find the possible lengths of the remaining side y.

2. A triangle has side 4 opposite 53° and another side 3. Find its area.

1

$$\sin A = \frac{10 \sin 61^\circ}{9}$$

$$A = 76.36\ldots^\circ \text{ or } 103.63\ldots^\circ$$

$$B = 180^\circ - 61^\circ - A$$

$$y = \frac{9 \sin B}{\sin 61^\circ}$$

$$y = 6.97 \text{ or } 2.73 \text{ (3 s.f.)}$$

2

$$\sin B = \frac{3 \sin 53^\circ}{4}$$

$$B = 36.79\ldots^\circ$$

$$C = 180^\circ - 53^\circ - B$$

$$\text{Area} = \frac{1}{2}(3)(4) \sin C$$

$$= 6.00 \text{ (3 s.f.)}$$

Problem Solving With Sine/Cosine Rule — Solution 37

Four mobile phone masts form convex quadrilateral ABCD. BC=75 m, CD=80 m, angle BCD=55° and angle ADC=140°. Masts must be at least 70 m apart, and AD is the minimum permitted distance.

a. Find AB.

a

$$BD^2 = 75^2 + 80^2 - 2(75)(80)\cos 55^\circ$$
$$BD = 71.708\ldots$$
$$\angle BDC = \sin^{-1}\left(\frac{75\sin 55^\circ}{BD}\right)$$
$$= 58.954\ldots^\circ$$
$$\angle BDA = 140^\circ - \angle BDC$$
$$AB = \sqrt{70^2 + BD^2 - 140BD\cos\angle BDA}$$
$$AB = 92.1 \text{ m}$$

b. Find angle BAD.

b

$$\sin\angle BAD = \frac{BD\sin\angle BDA}{AB}$$
$$\angle BAD = 50.3^\circ$$

c. Find the enclosed area.

c

$$\text{Area} = \frac{1}{2}(75)(80)\sin 55^\circ + \frac{1}{2}(70)(BD)\sin\angle BDA$$
$$= 4940 \text{ m}^2 \text{ (3 s.f.)}$$

Exercise 9E – Y1 page 189-191 — Solution 38

12. Triangle ABC has perimeter 15 cm, AB=7 cm and angle BAC=60°. Find AC, BC and the area.

Sides. Let AC=x, so BC=8−x.

Sides

$$(8-x)^2 = 7^2 + x^2 - 14x\cos 60^\circ$$
$$64 - 16x = 49 - 7x$$
$$x = \frac{5}{3}$$
$$AC = \frac{5}{3}\text{ cm}, \quad BC = \frac{19}{3}\text{ cm}$$

Area. Use the included angle.

Area

$$\text{Area} = \frac{1}{2}(7)\left(\frac{5}{3}\right)\sin 60^\circ$$
$$= \frac{35\sqrt{3}}{12}\text{ cm}^2 \approx 5.05\text{ cm}^2$$

Exercise 9E – Y1 page 189-191 Solution 39

13. Triangle ABC has AB=14 cm, BC=12 cm and CA=15 cm.

a. Find angle C to 3 significant figures.

a

$$\cos C = \frac{12^2 + 15^2 - 14^2}{2(12)(15)}$$

$$C = 61.3^\circ \text{ (3 s.f.)}$$

b. Find the area to 3 significant figures.

b

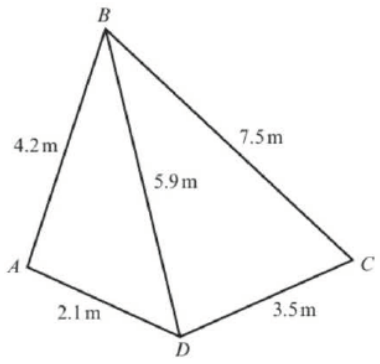
$$\text{Area} = \frac{1}{2}(12)(15)\sin C$$

$$= 78.9 \text{ cm}^2$$

Keep the unrounded angle in the calculation.

Exercise 9E – Y1 page 189-191 Solution 40

14. A flower bed is the quadrilateral shown. All lengths are in metres.



a. Find angles DAB and BCD.

a

$$\cos A = \frac{2.1^2 + 4.2^2 - 5.9^2}{2(2.1)(4.2)}$$

$$A = 136.3^\circ$$

$$\cos C = \frac{3.5^2 + 7.5^2 - 5.9^2}{2(3.5)(7.5)}$$

$$C = 50.1^\circ$$

b. Find the total area.

b

$$\frac{1}{2}(2.1)(4.2)\sin A + \frac{1}{2}(3.5)(7.5)\sin C$$

$$= 13.1 \text{ m}^2$$

c. Find diagonal AC.

c

$$\angle ADB = 29.4408\ldots^\circ$$

$$\angle BDC = 102.8569\ldots^\circ$$

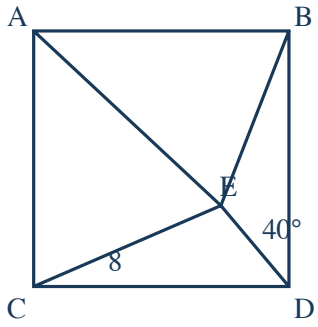
$$\angle ADC = 132.2978\ldots^\circ$$

$$AC^2 = 2.1^2 + 3.5^2 - 2(2.1)(3.5)\cos \angle ADC$$

$$AC = 5.15 \text{ m}$$

Exercise 9E – Y1 page 189-191 Solution 41

15. ABCD is a square of side 10 cm, with C bottom-left and D bottom-right. E lies inside the square, CE=8 cm and angle BDE=40°. Angle CED is obtuse. Find the area of triangle ABE.



1. Find the height of E above CD.

1

$\angle CDE = 50^\circ$

$\sin \angle CED = \dfrac{10 \sin 50^\circ}{8}$

$\angle CED = 106.753\ldots^\circ$

$\angle ECD = 23.246\ldots^\circ$

$h = 8 \sin \angle ECD = 3.157\ldots$ cm

2. Find the area of triangle ABE.

2

Area $ABE = \dfrac{1}{2}(10)(10-h)$

$= 34.2$ cm^2 (3 s.f.)

Extension Problem 1 Solution 42

[AEA 2009 Q5(a)] Triangle ABC has sides BC=a, AC=b, AB=c, with a<b<c. Angles A, B, C form an arithmetic sequence. Given $a = 2$ and $\sin A = \sqrt{15}/5$:

i. Show that the area is $ac\sqrt{3}/4$.

i

The middle angle B is 60° because the three angles sum to 180°.

Area $= \dfrac{1}{2} ac \sin 60^\circ = \dfrac{ac\sqrt{3}}{4}$

ii. Find b.

ii

$b = \dfrac{a \sin 60^\circ}{\sin A} = \sqrt{5}$

iii. Find c.

iii

Since A<60°,

$\cos A = \sqrt{1 - 3/5} = \dfrac{\sqrt{10}}{5}$

$c^2 - 2c + 4 = 5$

$c = 1 + \sqrt{2}$

Reject $1-\sqrt{2}$ because it is negative.

Extension Problem 2

Solution 43

[STEP I 2006 Q8] O=(0,0,0), A=(a,0,0), B=(0,b,0), C=(0,0,c), where a,b,c>0. Let angle ACB=θ and let d be the perpendicular distance from O to plane ABC.

i. Find the volume of tetrahedron OABC.

i

$$V = \frac{1}{3}\left(\frac{1}{2}ab\right)c = \frac{1}{6}abc$$

ii. Find cos θ and the area Δ of ABC.

ii

$$AB^2 = a^2 + b^2,\ AC^2 = a^2 + c^2$$

$$BC^2 = b^2 + c^2$$

$$\cos\theta = \frac{c^2}{\sqrt{(a^2+c^2)(b^2+c^2)}}$$

$$\Delta = \frac{1}{2}AC \cdot BC\sin\theta$$

$$= \frac{1}{2}\sqrt{a^2b^2 + b^2c^2 + c^2a^2}$$

iii. Show that $d^{-2} = a^{-2} + b^{-2} + c^{-2}$.

iii

$$\frac{1}{3}\Delta d = \frac{1}{6}abc$$

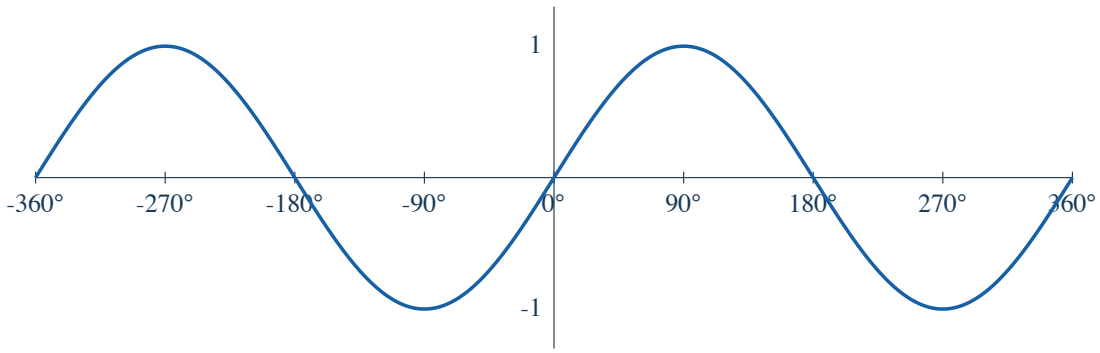
$$\frac{1}{d^2} = \frac{4\Delta^2}{a^2b^2c^2}$$

$$= \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}$$

Sin Graph

Solution 44

$y = \sin x$



Features. State the period, range and zeros.

Features

Period $= 360°$, $\quad -1 \le \sin x \le 1$

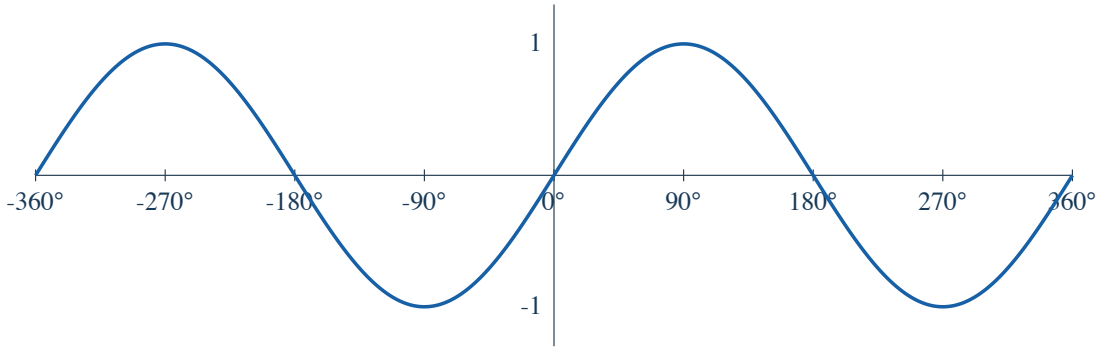
Zeros: $x = 180°n,\ n \in \mathbb{Z}$

Maximum 1 at 90°+360°n; minimum −1 at 270°+360°n.

Sin Graph

Solution 45

Given $\sin 30^\circ = \frac{1}{2}$, use symmetry of the sine graph.



a.

$\sin 150^\circ$

a

$\sin(180^\circ - 30^\circ) = \frac{1}{2}$

b.

$\sin(-30^\circ)$

b

$-\sin 30^\circ = -\frac{1}{2}$

c.

$\sin 210^\circ$

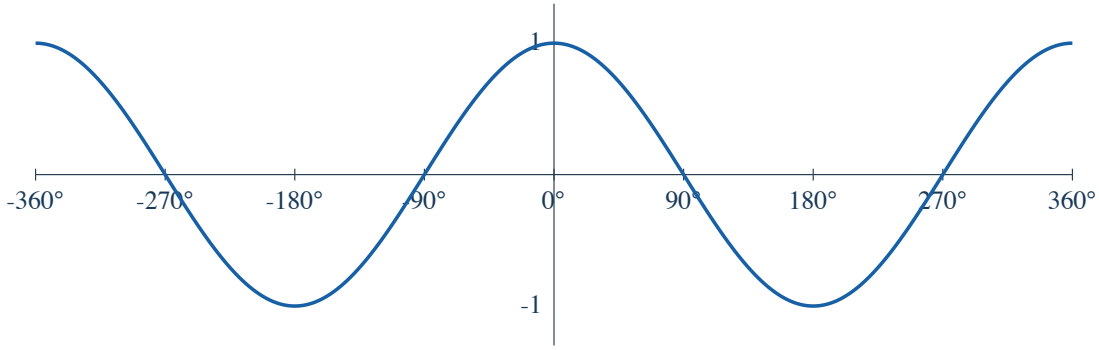
c

$\sin(180^\circ + 30^\circ) = -\frac{1}{2}$

Cos Graph

Solution 46

$y = \cos x$



Features. State the period, range and zeros.

Features

Period $= 360^\circ$, $\quad -1 \leq \cos x \leq 1$

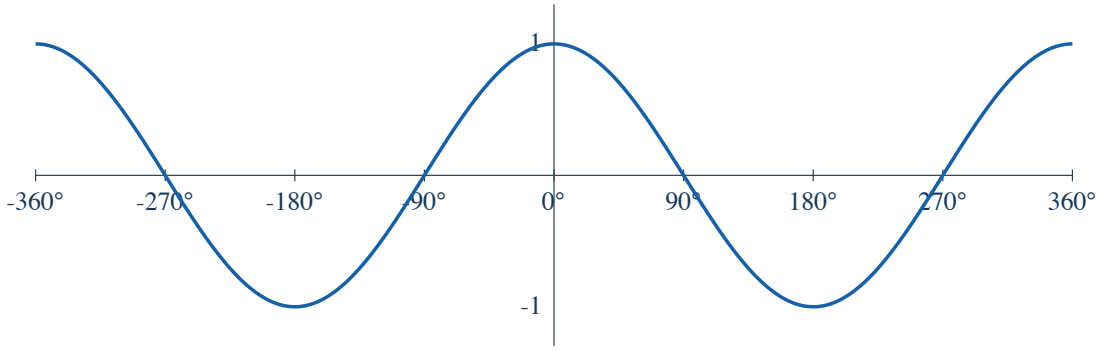
Zeros: $x = 90^\circ + 180^\circ n,\ n \in \mathbb{Z}$

Maximum 1 at 360°n; minimum -1 at $180^\circ + 360^\circ$n.

Cos Graph

Solution 47

Given $\cos 60^\circ = \frac{1}{2}$, use symmetry of the cosine graph.



a.

$\cos 120^\circ$

a

$\cos(180^\circ - 60^\circ) = -\frac{1}{2}$

b.

$\cos(-60^\circ)$

b

$\cos 60^\circ = \frac{1}{2}$

c.

$\cos 240^\circ$

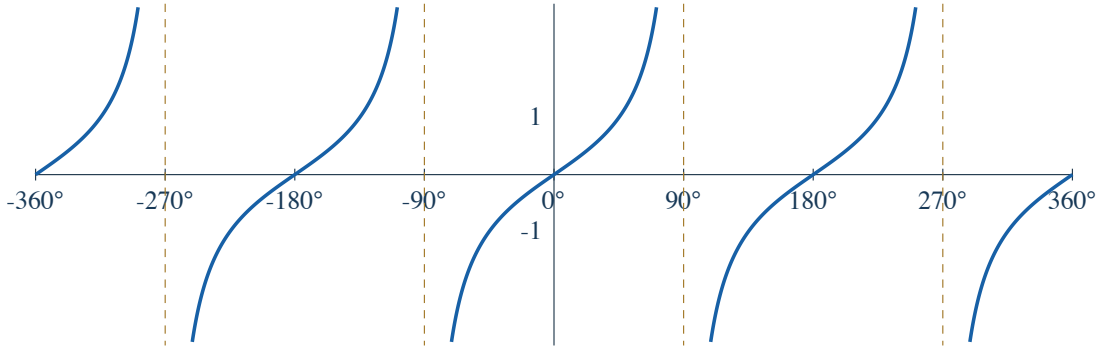
c

$\cos(180^\circ + 60^\circ) = -\frac{1}{2}$

Tan Graph

Solution 48

$y = \tan x$



Features. State the period, range, zeros and asymptotes.

Features

Period $= 180^\circ$, range: all real values

Zeros: $x = 180^\circ n$

Vertical asymptotes: $x = 90^\circ + 180^\circ n$

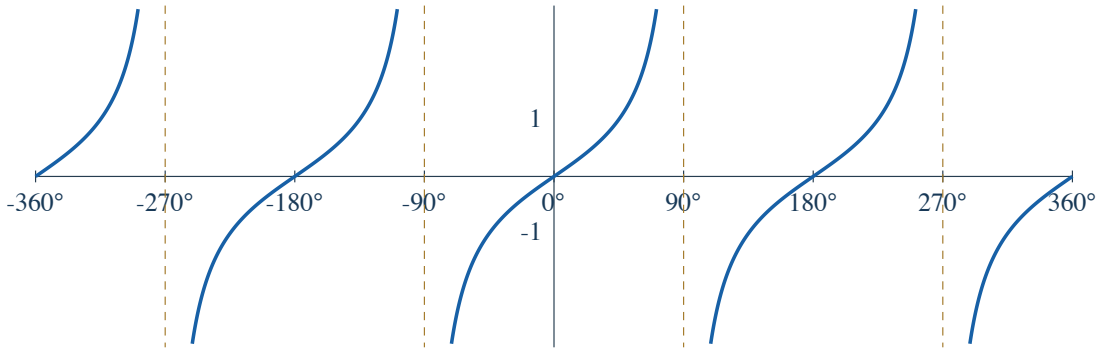
$n \in \mathbb{Z}$

Tangent is undefined at its vertical asymptotes.

Tan Graph

Solution 49

Given $\tan 30^\circ = 1/\sqrt{3}$, use symmetry of the tangent graph.



a.

$\tan(-30^\circ)$

a

$$-\tan 30^\circ = -\frac{1}{\sqrt{3}}$$

b.

$\tan 150^\circ$

b

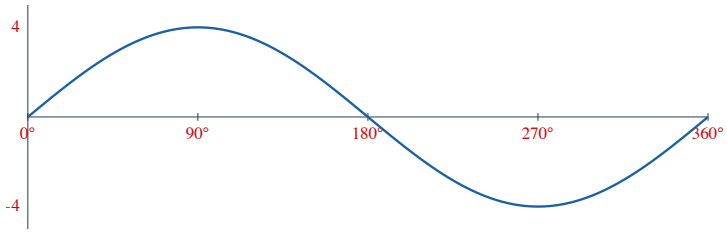
$$\tan(-30^\circ + 180^\circ) = -\frac{1}{\sqrt{3}}$$

Transforming Trigonometric Graphs

Solution 50

Use the same transformation rules as for other functions: changes outside affect outputs; changes inside affect inputs.

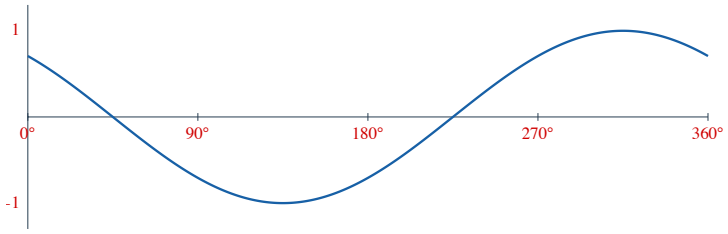
1. $y = 4\sin x, \quad 0^\circ \le x \le 360^\circ$



1 features

Vertical stretch by scale factor 4. Period remains 360˚; range $-4 \le y \le 4$.

2. $y = \cos(x + 45^\circ), \quad 0^\circ \le x \le 360^\circ$



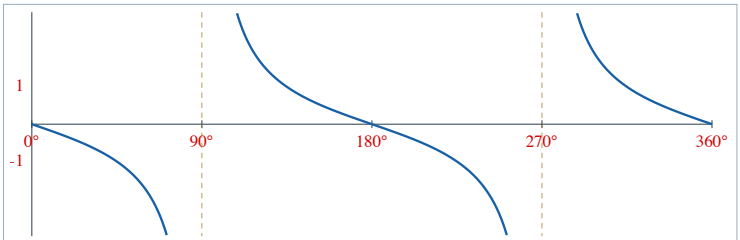
2 features

Translate 45˚ left. Zeros 45˚,225˚; minimum at 135˚; maximum at 315˚.

Transforming Trigonometric Graphs

Solution 51

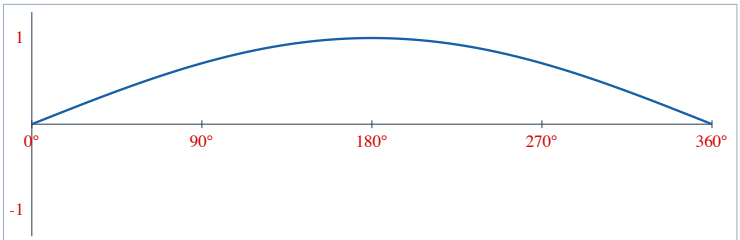
1. $y = -\tan x, \quad 0^\circ \leq x \leq 360^\circ$



1 features

Reflect in the x-axis. Zeros $0^\circ, 180^\circ, 360^\circ$; asymptotes $90^\circ, 270^\circ$. Each branch decreases.

2. $y = \sin(x/2), \quad 0^\circ \leq x \leq 360^\circ$



2 features

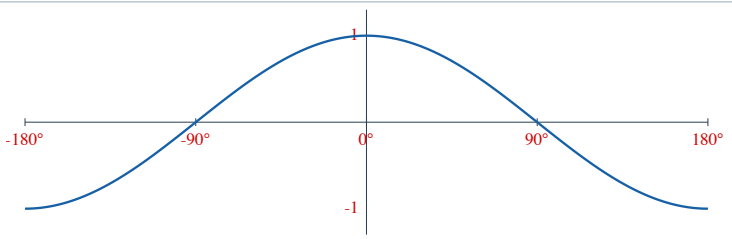
Horizontal stretch by scale factor 2. Period 720°. This interval shows half a cycle, with maximum $(180^\circ, 1)$.

Exercise 9F – Y1 Page 194

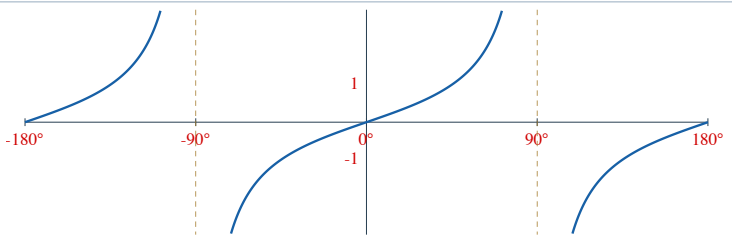
Solution 52

Sketch each graph on the stated interval.

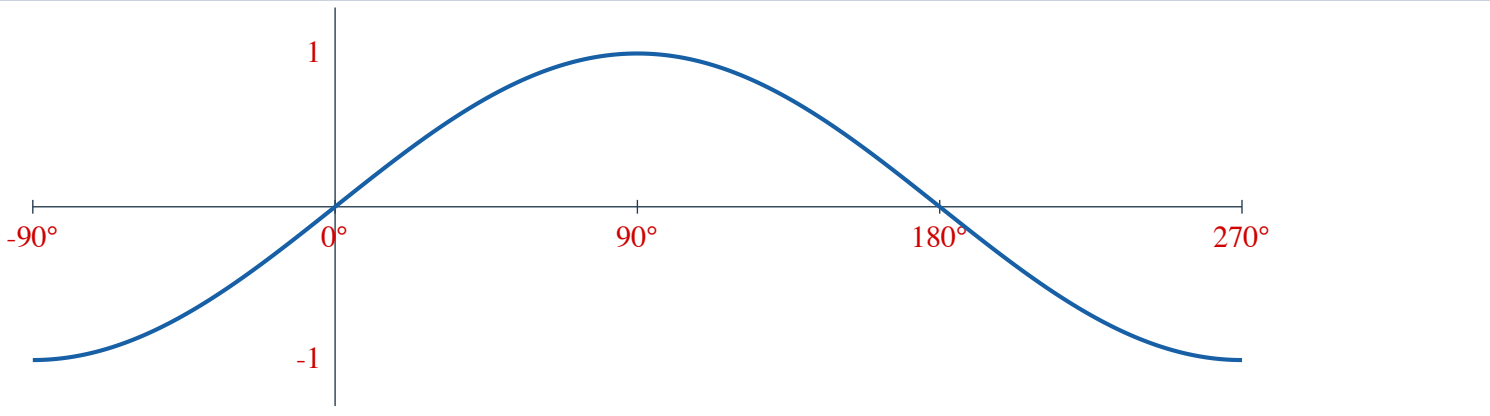
1. $y = \cos\theta, \ -180^\circ \leq \theta \leq 180^\circ$



2. $y = \tan\theta, \ -180^\circ \leq \theta \leq 180^\circ$

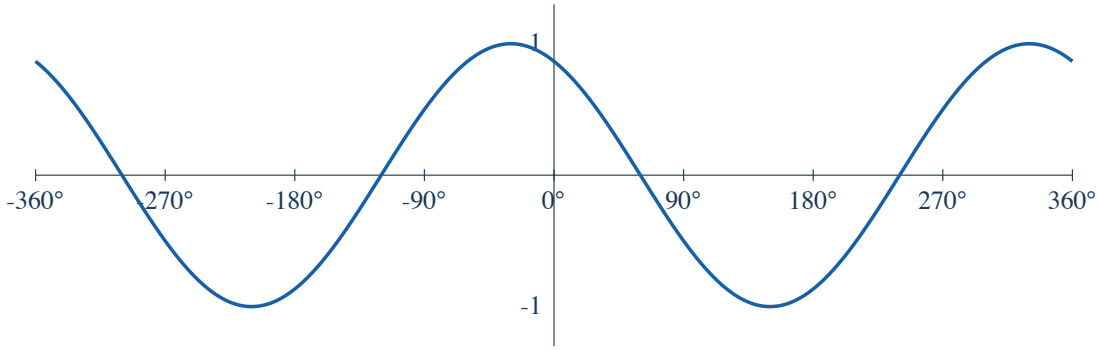


3. $y = \sin\theta, \ -90^\circ \leq \theta \leq 270^\circ$



Exercise 9G – Y1 Pages 197-198 Solution 53

7. $y = \cos(x + 30^\circ), \quad -360^\circ \leq x \leq 360^\circ$



a. Find the x-axis intersections.

a

$x + 30^\circ = 90^\circ + 180^\circ n$

$x = -300^\circ, -120^\circ, 60^\circ, 240^\circ$

Each corresponding coordinate is (x,0).

b. Find the y-axis intersection.

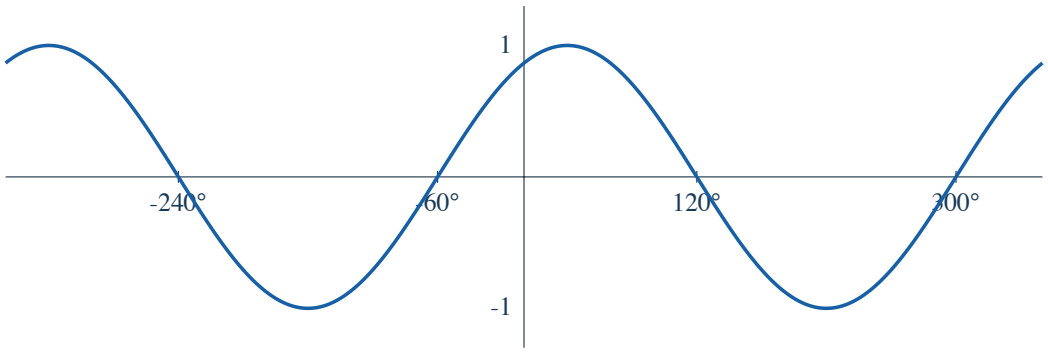
b

$y = \cos 30^\circ = \frac{\sqrt{3}}{2}$

$\left(0, \frac{\sqrt{3}}{2}\right)$

Exercise 9G – Y1 Pages 197-198 Solution 54

8. The graph has equation $y = \sin(x + k)$, where k is a constant (angles in degrees).



a. Give one possible value of k.

a

$k = 60^\circ$

The rising zero is at -60°.

b. Is there more than one possible value?

b

$k = 60^\circ + 360^\circ n, \quad n \in \mathbb{Z}$

Yes: sine repeats every 360°.

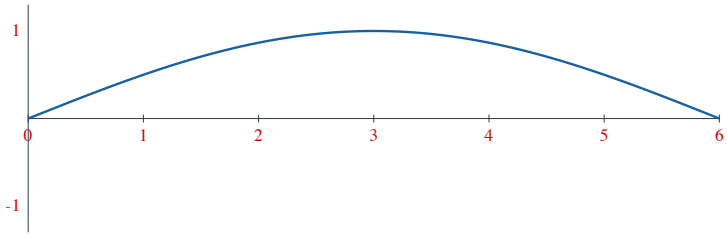
Exercise 9G – Y1 Pages 197-198 Solution 55

9. The relative water depth of a rock pool is modelled by $y = \sin(30t)^\circ$, where t is time in hours and $0 \leq t \leq 6$. t=0 represents midday.

a. Sketch the function.

b. During what times is the pool at least half full?

a



b

$\sin(30t)^\circ \geq \frac{1}{2}$

$30^\circ \leq 30t \leq 150^\circ$

$1 \leq t \leq 5$

From 1 pm to 5 pm, including both endpoints.

Extension Solution 56

1. [MAT 2013 1B] Reflect $y = \sin x$ first in x=π, then in y=2. Find the resulting equation.

2. [MAT 2011 1D] What fraction of $0^\circ \leq x \leq 360^\circ$ satisfies at least one of $\sin x \geq \frac{1}{2}$ or $\sin 2x \geq \frac{1}{2}$?

1

$y = \sin(2\pi - x) = -\sin x$

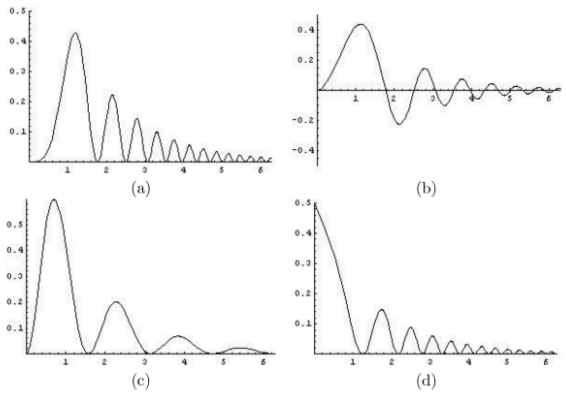
$y = 4 - (-\sin x) = 4 + \sin x$

2

First interval: $[30^\circ, 150^\circ]$. Second: $[15^\circ, 75^\circ]$ and $[195^\circ, 255^\circ]$. Their union has length $135^\circ + 60^\circ = 195^\circ$.

$\frac{195}{360} = \frac{13}{24}$

3. [MAT 2007 1G] Choose the sketch of $y = 2^{-x}\sin^2(x^2)$.

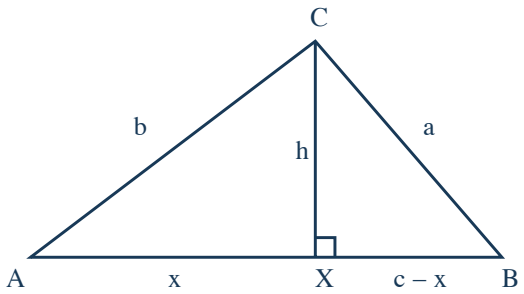


3

Answer (a). The function is non-negative, is zero at x=0, and has a decreasing envelope 2^{-x}. As x increases the oscillations become closer together.

APPENDIX :: Proof of Cosine Rule

Solution 57



Use Pythagoras in the two right-angled triangles.

$$h^2 + x^2 = b^2$$
$$h^2 + (c - x)^2 = a^2$$

Drop a perpendicular CX to AB. Let AX=x and AB=c.

2. Eliminate h^2 and expand.

2

$$a^2 - b^2 = (c - x)^2 - x^2$$
$$a^2 = b^2 + c^2 - 2cx$$

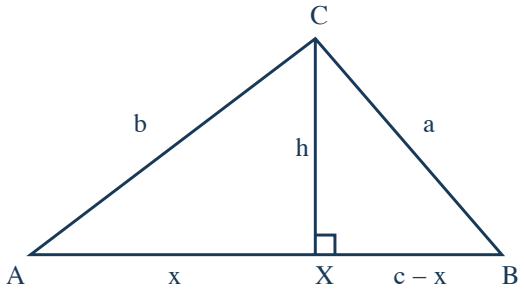
3. Use the angle at A.

3

$$x = b\cos A$$
$$a^2 = b^2 + c^2 - 2bc\cos A$$

APPENDIX :: Proof of Sine Rule

Solution 58



1. Express h in two ways.

1

$$h = b\sin A = a\sin B$$

The same altitude h belongs to both right-angled triangles.

2. Rearrange.

2

$$a\sin B = b\sin A$$
$$\frac{a}{\sin A} = \frac{b}{\sin B}$$

3. Repeat with another altitude.

3

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$